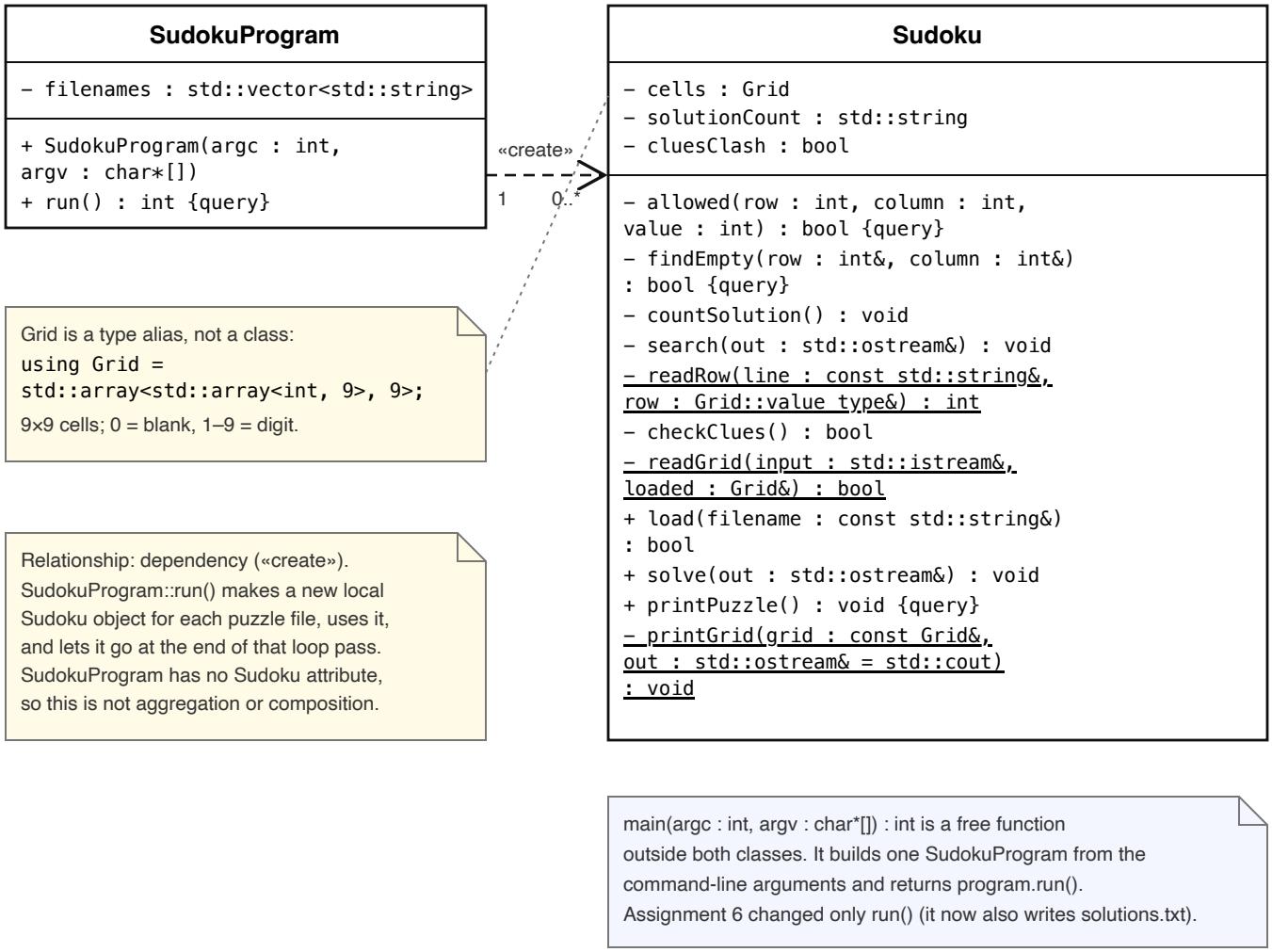
EECS 348 Assignment 6: UML Class Diagrams

Francis Tryban · Assignment6.cpp (modified from Luke Tidball's Assignment 5) · November 13, 2026
Diagrams generated with Claude Opus 5.5 (Anthropic, through Claude Code) directly from the final Assignment6.cpp.

Key: – private, + public. Attributes are `visibility name : type`. Methods are `visibility name(parameter : type, …) : return type`. Underlined = static. `{query}` = const method (does not change the object).



Class: SudokuProgram

Member	Visibility	Kind	Purpose
filenames : std::vector<std::string>	private	attribute	Puzzle files to solve, in order (puzzle1–5.txt, or the command-line names).
SudokuProgram(argc : int, argv : char*[])	public	constructor	Fills `filenames` from the arguments, or the five defaults.

run() : int {query}	public	method	Opens solutions.txt once, then for each file: loads and solves a Sudoku, and writes the results to solutionN.txt and solutions.txt. Returns the exit status.

Class: Sudoku

Member	Visibility	Kind	Purpose
cells : Grid	private	attribute	The 9×9 puzzle being solved (0 = blank).
solutionCount : std::string	private	attribute	Number of solutions found, stored as decimal text.
cluesClash : bool	private	attribute	True when the starting clues break a Sudoku rule.
allowed(row, column, value : int) : bool {query}	private	method	Checks the row, column and 3×3 box for a repeat.
findEmpty(row : int&, column : int&) : bool {query}	private	method	Finds the next blank cell.
countSolution() : void	private	method	Adds one to `solutionCount`.
search(out : std::ostream&) : void	private	method	Recursive backtracking; writes each finished grid to `out`.
readRow(line : const std::string&, row : Grid::value_type&) : int	private, static	method	Reads one line of cells; returns how many were read.
checkClues() : bool	private	method	Checks that the given clues do not conflict.
readGrid(input : std::istream&, loaded : Grid&) : bool	private, static	method	Reads nine valid rows into a grid.
load(filename : const std::string&) : bool	public	method	Opens and reads a puzzle file; false if it cannot.
solve(out : std::ostream&) : void	public	method	Runs the search and writes "Solutions found: n" or "No solution found."
printPuzzle() : void {query}	public	method	Prints the starting grid (kept from Assignment 4; not called).
printGrid(grid : const Grid&, out : std::ostream& = std::cout) : void	private, static	method	Writes a grid in the input-file format, then a blank line.